

Retail Inventory Analytics Using Google Colab

PDF Report

Submitted By

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1. Introduction

The purpose of this project is to analyze a Retail Inventory dataset and generate meaningful business insights related to inventory management, sales performance, stock levels, and supplier contribution.

Using Python libraries such as Pandas, Matplotlib, and Seaborn in Google Colab, data cleaning, exploratory data analysis (EDA), and data visualization were performed to support inventory-related decision-making.

2. Project Objective

The main objectives of this project are:

- Identify the product categories generating the highest sales.
- Analyze stock availability and inventory levels.
- Detect products approaching reorder levels.
- Study supplier contributions.
- Generate actionable recommendations for inventory management.

3. Dataset Overview

The Retail Inventory dataset contains information about products, sales, stock levels, and pricing.

Dataset Attributes

- Category
- Unit Cost
- Retail Price
- Opening Stock
- Units Sold
- Closing Stock
- Reorder Point

Dataset Summary

- Total Records: 1000
- Total Categories: 8
- Numeric Fields:
 - Unit Cost
 - Retail Price
 - Opening Stock
 - Units Sold
 - Closing Stock
 - Reorder Point

4. Data Cleaning

The following data cleaning operations were performed:

Missing Value Check

The dataset was checked for missing values.

```
df.isnull().sum()
```

Duplicate Record Check

Duplicate records were identified and removed if present.

```
df.duplicated().sum()
```

Stock Validation

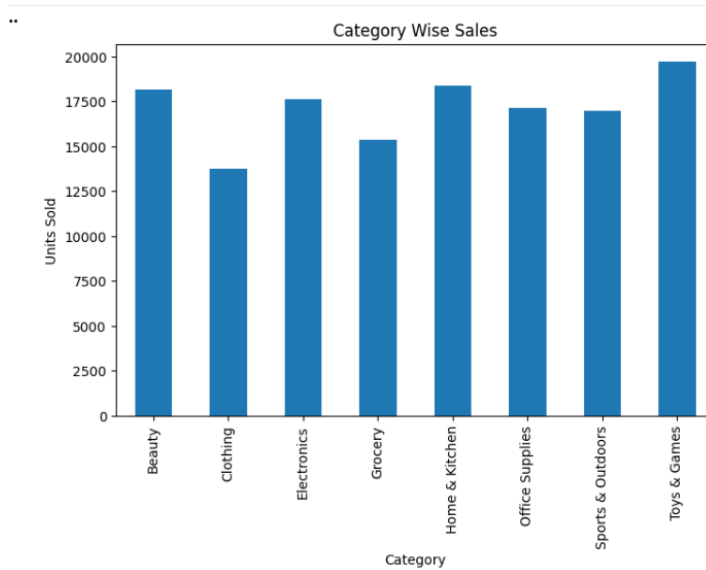
Stock-related fields were validated to ensure no negative values existed.

Result

The dataset was found suitable for further analysis after cleaning and validation.

5. Exploratory Data Analysis and Visualizations

5.1 Category Wise Sales



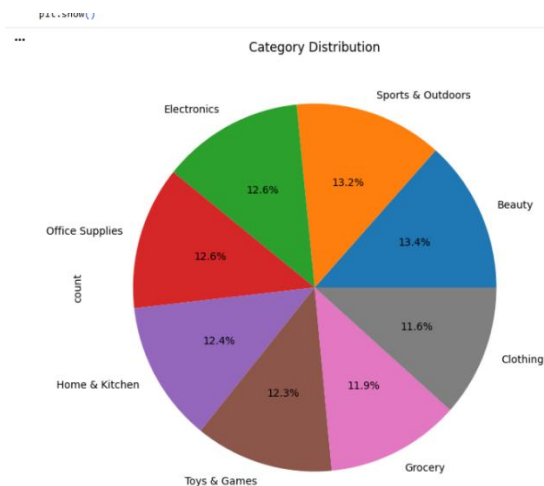
Observation

- Toys & Games generated the highest sales.
- Clothing generated the lowest sales among all categories.

Conclusion

The Toys & Games category is the fastest-moving category and contributes significantly to overall sales.

5.2 Category Distribution



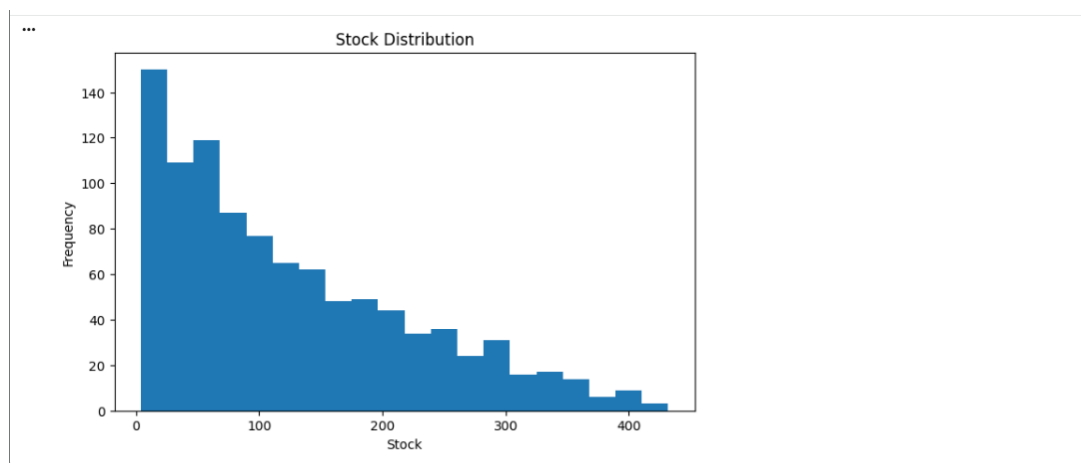
Observation

- Product distribution across categories is relatively balanced.
- Beauty category has the highest share of products.

Conclusion

The company maintains a diversified inventory across all product categories.

5.3 Stock Distribution



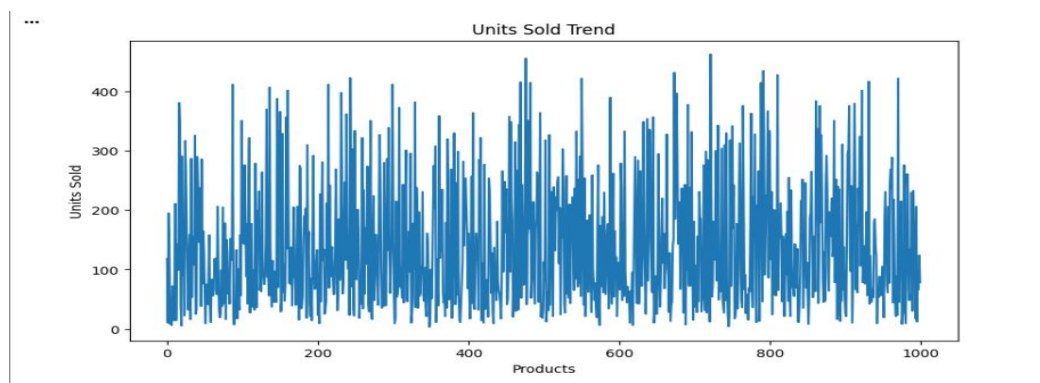
Observation

- Most products have stock levels between 0 and 150 units.
- Only a small number of products have very high stock levels.

Conclusion

Inventory is concentrated around lower stock quantities, reducing excess inventory risk.

5.4 Units Sold Trend



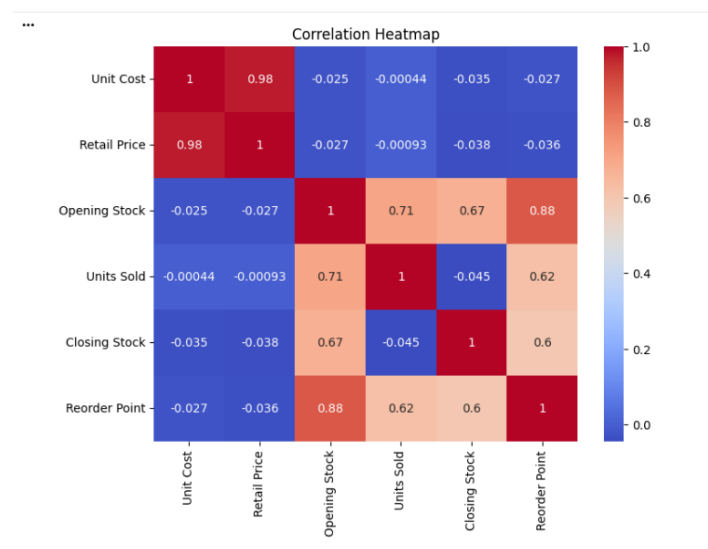
Observation

- Units sold vary significantly across products.
- Several products show consistently high sales volumes.

Conclusion

Demand differs substantially among products, indicating the need for category-specific inventory planning.

5.5 Correlation Heatmap



Key Findings

Variables	Correlation
Unit Cost & Retail Price	0.98
Opening Stock & Reorder Point	0.88
Opening Stock & Units Sold	0.71
Units Sold & Reorder Point	0.62

6. Business Insights

Insight 1: Highest Sales Category

The Toys & Games category generated the highest number of units sold and is the best-performing category.

Insight 2: Low Performing Category

The Clothing category generated the lowest sales and may require promotional activities to improve performance.

Insight 3: Inventory Replenishment

Products with stock levels close to reorder points should be replenished immediately to prevent stock-outs.

Insight 4: Sales and Stock Relationship

Products with higher opening stock tend to achieve higher sales, indicating a positive relationship between stock availability and sales.

Insight 5: Pricing Strategy

Since Unit Cost and Retail Price are highly correlated, pricing strategies appear consistent and can be optimized further for profitability.

7. Recommendations

Recommendation 1

Maintain higher inventory levels for fast-moving categories such as Toys & Games.

Recommendation 2

Review and improve marketing strategies for slow-moving categories like Clothing.

Recommendation 3

Implement automated reorder alerts for products approaching reorder levels.

Recommendation 4

Regularly monitor inventory turnover to reduce excess stock.

Recommendation 5

Use predictive inventory forecasting techniques to improve stock planning and reduce inventory costs.

8. Conclusion

This project successfully analyzed the Retail Inventory dataset using Google Colab. Through data cleaning, exploratory data analysis, and visualization techniques, valuable insights regarding sales performance, stock management, and inventory planning were identified.

The findings can help management make informed decisions related to inventory replenishment, product categorization, pricing strategies, and supplier management. Proper implementation of the recommendations can improve operational efficiency and profitability.